

Week 11 unsupervised learning

1. How can we generate similar data using discrete and continuous random variables

In unsupervised learning, it is often useful to generate similar data to the original dataset in order to augment the data or to simulate new scenarios for further analysis. Discrete and continuous random variables can be used to generate similar data in unsupervised learning in the following ways:

Discrete Random Variables: Suppose the original dataset contains a categorical variable with a finite number of possible values. One way to generate similar data is to use a discrete random variable to generate samples from the same distribution as the original variable. This can be done by estimating the probability distribution of the original variable from the data, and then using this distribution to generate new samples. One popular method for doing this is to use the multinomial distribution, which models the probability of each category given the observed frequencies in the data.

Continuous Random Variables: Suppose the original dataset contains a continuous variable. One way to generate similar data is to use a continuous random variable to generate samples from the same distribution as the original variable. This can be done by estimating the probability distribution of the original variable from the data, and then using this distribution to generate new samples. One popular method for doing this is to use the Gaussian distribution, which **models the probability density of the variable as a bell-shaped curve with a mean and standard deviation estimated from the data.**

Both discrete and continuous random variables can also be used in combination to generate data with multiple variables. For example, if the original dataset contains both categorical and continuous variables, a joint probability distribution can be estimated and used to generate new samples with similar characteristics.

In summary, discrete and continuous random variables can be used to generate similar data in unsupervised learning by estimating the probability distribution of the original data and then using this distribution to generate new samples.

Density function (probability density function, PDF)

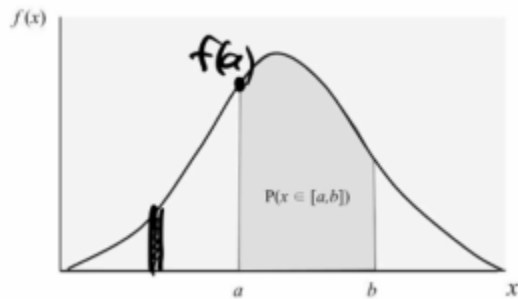
(probability problem: the probability of a bubble bursting X is in any time from 0 seconds to 1 minute. The probability of the bubble bursting at a particular value X is instantly small. We should base the probability on when bubbles normally burst from a probability distribution $f(x)$.)

A density function $f(x)$ represents the probability distribution of a continuous variable X . (For example X is the time when a bubble burst). The probability of X being in the interval $[a, b]$ is given by the area under the curve of $f(x)$ over that interval.

In the context of the bubble bursting, if we have a density function $f(x)$ that represents the probability distribution of the time it takes for a bubble to burst, then the probability of the bubble bursting in the interval $[a, b]$ is given by the area under the curve $f(x)$ over that interval. The area represents the probability that the bubble will burst within that time range.

PDFs are utilized in modeling the underlying distribution, constructing generative models, sampling new data points, and evaluating the similarity between the generated and training data distributions. (int the context of learning how to generate new data)

a density function $f(x)$ where the probability $P(X \in [a, b])$ (X is an element between a and b) = area under the curve from f on interval $[a, b]$.



Density function in 1D and in 2D.

- 1D: the probability of where a point will appear on the line is based on the distribution of the points. Comparing the area of the points we can decide the probability of where a new point will appear.
- 2D: circles are peaks, some are steeper than others.

Må ha lik probability distrubution på ny data.

2. What is hierarchical clustering?

Hierarchical clustering is a way to group the data. Initially, each sample is a cluster. Calculate the distance between every pair of clusters. Find the two clusters with smallest distance and merge them together. Repeat until we have one cluster or the desired number of clusters.

We have to define the distance between two clusters: single linkage, average linkage, complete linkage and ward linkage.

Single linkage: find the point nearest the other cluster and find distance. Smallest distance between the two clusters.

Average linkage: average distance between samples in c_1 and c_2

Complete linkage: maximum distance between a sample in c_1 and c_2

Ward linkage: increase within-cluster variance if clusters are merged.

3. What is k-means clustering?

k-means is a way to group data. Randomly select k cluster centers. Assign each sample to nearest cluster center. Update cluster center to mean of samples in cluster. Repeat assigning each sample to a cluster and updating the centers.

New center is updated by calculating the average of all sample vectors or median

$$c_i^{new} = \frac{1}{m} \sum x_i$$

4. What is k-means neural network?

A k-means neural network is a neural network architecture that combines the k-means clustering algorithm with artificial neural networks. This type of network uses the k-means algorithm to cluster the input data into k clusters, and then trains a neural network on the clustered data.

The k-means neural network can be seen as a two-stage process:

1. In the first stage, the k-means algorithm is applied to the input data to form k clusters.
2. In the second stage, a neural network is trained on the clustered data to perform a specific task, such as classification or regression.

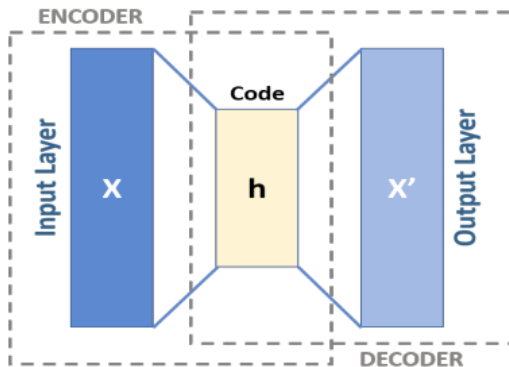
The advantage of using a k-means neural network is that it can reduce the complexity of the input data by clustering similar data points together, which can improve the performance and efficiency of the neural network. Additionally, the use of clustering can make the neural network more interpretable, as it can provide insights into the structure of the input data.

5. what is autoencoders?

Autoencoder is a way to compress the data. It produces an output similar to the input. I have as many output nodes as input nodes. It forces the data to compress by adding a hidden layer with fewer nodes. We need a loss function, and we often use reconstruction loss

Fjerne noise. Ting som er en rar observasjon.

$$Loss(W) = \sum (x_i - output(x_i))^2$$



6. Why would we want to reduce the number of features in a dataset?

We want to reduce the number of features to easier visualize the data, remove excess noise, or to reduce number of features before further analysis.

7. What is the difference between compression and reduction?

Compression is not for human interpretation, and should be possible to decode to original format.

Reduction is for human interpretation, should only preserve important structure. Decoding is not an aim.

8. What is pca? How can we find eigenvector and eigenvalue?

Pca reduces the number of features. It works by finding the line or vector that best fit the data

PCA (Principal Component Analysis) is a dimensionality reduction technique that is used to reduce the number of input features or variables while retaining the most important information in the data. PCA works by transforming the original input data into a new coordinate system, where the first axis, known as the first principal component, captures the maximum amount of variation in the data.

Here are the steps involved in performing PCA:

1. Standardize or center the data: PCA requires that the input data is standardized so that each variable has zero mean and unit variance. This step ensures that each variable is equally important in the analysis.
2. Calculate the covariance matrix: The covariance matrix measures how much two variables vary together. The diagonal elements of the covariance matrix represent the variance of each variable, while the off-diagonal elements represent the covariance between variables.
3. Calculate the eigenvectors and eigenvalues: The eigenvectors and eigenvalues are calculated from the covariance matrix. The eigenvectors represent the directions of maximum variation in the data, while the eigenvalues represent the amount of variation captured by each eigenvector.

4. Choose the principal components: The eigenvectors are sorted in descending order based on their corresponding eigenvalues, and the first k eigenvectors are selected as the principal components.

5. Transform the data: The original input data is transformed into a new coordinate system using the selected principal components. Dot produktet mellom original data og egenvektor

9. What happens when we consider two or more eigenvectors?

if we consider the second eigenvector, we would be including the second most important (or significant) source of variation in the data. This can be useful if we want to capture a different variation that the first vector did not show.

The results would have been different. The distance between the points would have been smaller, since we do not choose the optimal dimension that maximizes the distance between the points.

If we consider both eigenvectors, we would have included all the sources of variation in the data. This would have been the same as the original dataset, without performing pca since we only have two dimensions to start with.

If we had more dimensions than two, then it could benefit to consider both the first and second eigenvectors. Then we would consider the two most significant sources of variation in the data. This would provide us with a more complete picture of the variation in the data and could help us to identify more complex patterns and relationships. However, including both eigenvectors could also result in overfitting if the data is noisy or if there are too many features.